

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

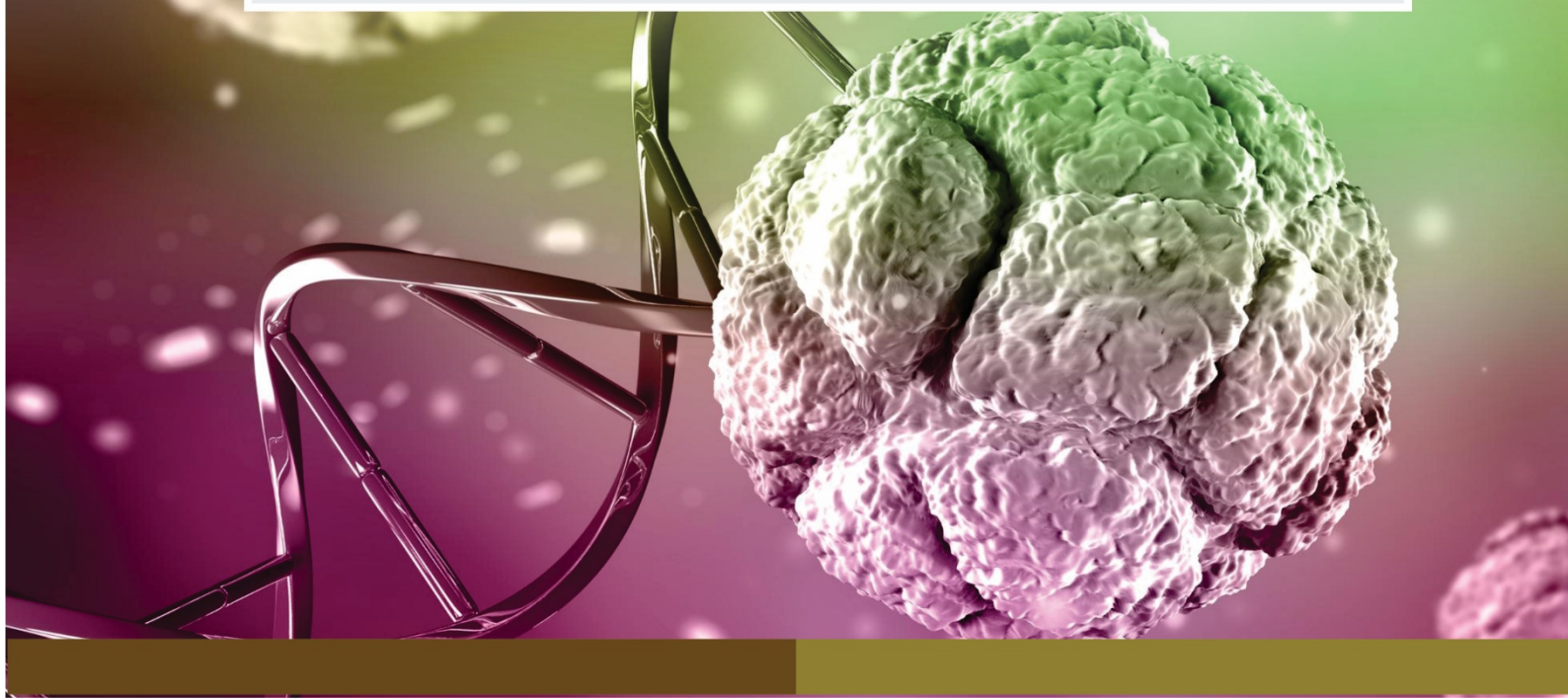
BIOLOGY

Topic wise
Notes

LECTURE

15

Circulatory System, Part - 01



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Circulatory System

Group of organs involved in transportation of materials (nutrients, hormone, respiratory gases) excretory material etc. is known as circulatory System.

- Branch of Science deals with study of circulatory systems known as **Angiology**.
- Father of Angiology known as **William Harvey**

Components of Circulatory Systems—

1. Human Heart
2. Blood
3. Blood Vessels

Circulatory Pathways

There are two types of circulatory systems:

1. Open circulatory System
2. Closed circulatory System

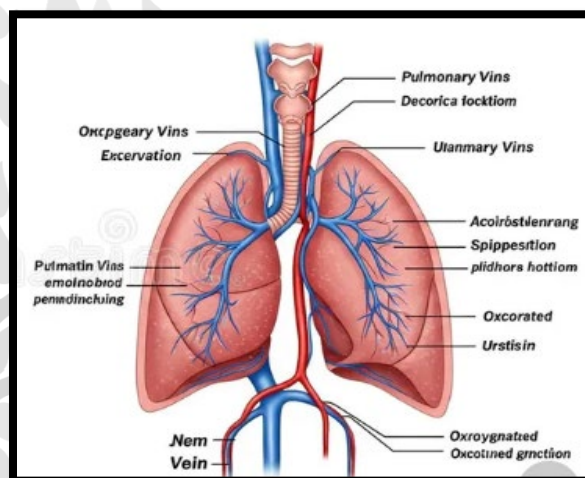
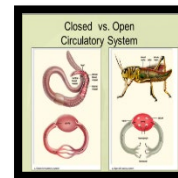
Open type	Closed type
(1) In this blood is pumped by heart passes through large vessels into large open spaces or body cavities called blood sinuses.	(1) Blood is pumped by heart into closed network or blood vessels
(2) Tissues are in direct contact with circulating fluid.	(2) Tissues are not in direct contact with circulating fluid
(3) Less advantageous pattern eg. Arthropods, molluscs	(3) More advantageous pattern eg. Annelids, Chordates, (Class cephalopoda of molluscs)

1. **Open circulatory system:** Blood vessels are not found, and blood is present in the open cavity known as sinuses, where internal organs float. This type of system is present in arthropods and mollusks.
2. **Closed circulatory system:** Blood vessels circulate the blood. This type of system is present in Annelids and Chordates.

Name	Types of Heart	Circulation
Fish	2 chambered, 1-atrium and 1-ventricle	Single circulation
Amphibian and Reptiles	3 chambered, 2-atrium and 1-ventricle	Incomplete double circulation
Crocodiles, Birds and Mammals	4 chambered, 2-atrium and 2-ventricle	Double circulation

Human Circulatory System:

Heart- Four chambered, derived from mesoderm. It is present between the two lungs in the thoracic cavity and protected by the pericardium.



- Size: roughly a fist

Structure of Heart:

Heart the mesodermally derived organ, is situated in the thoracic cavity in between the lungs, slightly tilted to the left.

- Heart has size of clenched fist, weight 300 gm. Its triangular superior-broad portion is tilted slightly towards right (dorsal) side. Its lower narrow portion is tilted towards left side.
- Heart is protected by a double layered bag called pericardium. The narrow space in between these two membranes is called pericardial cavity in which pericardial fluid is present. Pericardial fluid provide moisture to heart and reduces frictions.
- It is secreted by the pericardium. Pericardial cavity is a true coelom (as it lies between two layers of mesoderm).





Wall of Heart : The wall of heart is made of three layers.

- (i) **Epicardium** – outermost layer, Made of simple squamous epithelium.
- (ii) **Myocardium** – middle layer, thickest, Made of cardiac muscles which are striated but involuntary .
- (iii) **Endocardium** – innermost layer, Made of simple squamous epithelium

Thickness of wall of Heart depends on Myocardium.

- The heart of man is four chambered. Two relatively small upper chambers called Atria and two larger lower chambers called ventricles

Systemic heart:

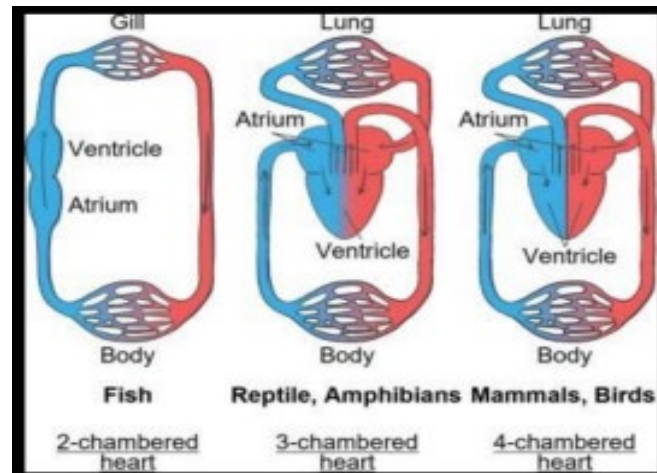
- left part of the heart (i.e. left atrium and left ventricle) contain the blood which is to be pumped into the Systemic circulation, therefore it is called systemic heart. The main purpose of such a circulation is to transport oxygen ,as well as nutrients to the body tissues, and to remove carbondioxide and other harmful nitrogenous waste from them.

Pulmonary heart:

- Right part of the heart (i.e. right atrium and right ventricle) contain the blood which is to be pumped in
- pulmonary circulation for oxygenation, therefore it is called pulmonary heart. The pulmonary circulation is responsible for regular oxygenation of the impure deoxygenated blood which is received by the right auricle.

4 Chambers:

- Right Atrium
- Right Ventricle
- Left Atrium
- Left Ventricle

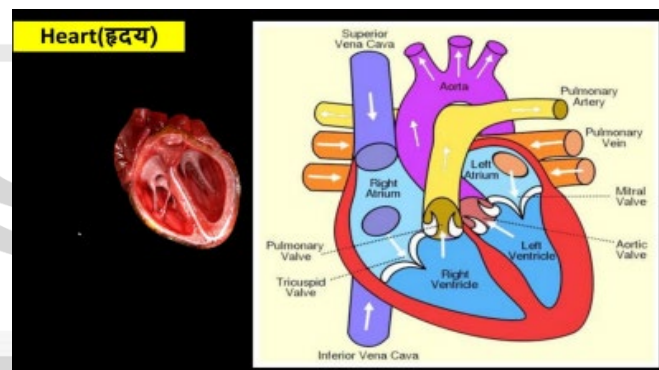


Valves (prevent backflow):

- Tricuspid Valve
- Bicuspid (Mitral) Valve
- Pulmonary Valve
- Aortic Valve

Function of Heart

- Pumps deoxygenated blood → lungs
- Pumps oxygenated blood → body
- Transportation of substances
- Regulation of blood flow
- Separation of oxygenated and deoxygenated



blood

- Generation of heartbeat
- Removal of waste product

Facts related to Heart:

- Mesodermal in origin
- First organ developed in fetus
- Myogenic in nature (Self heartbeat generation)





- Average Weight – 300 gm
- Location: Present in thoracic cavity kept in between lungs where its lower part is titled towards left side and its upper part is slightly titled towards right side.
- Position of heart is termed as mediastinal.
- Double membrane covering nearby known as pericardium.
- Fluid is present in pericardial cavity known as pericardial fluid.

